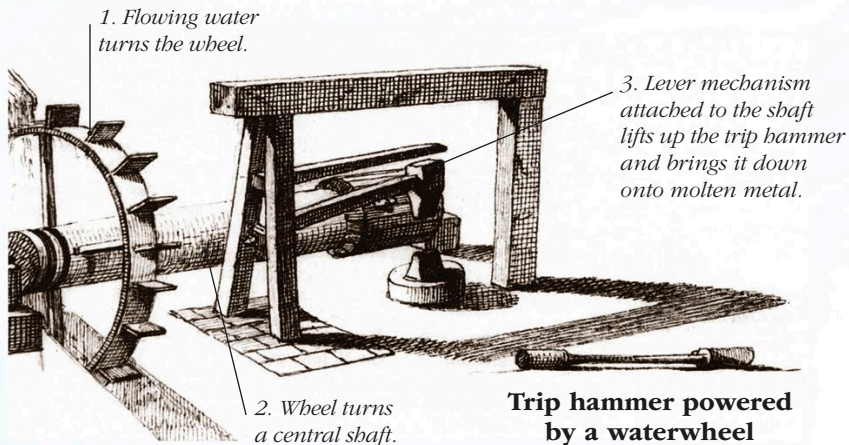


TRIP HAMMER

By the 1st century BCE, the ancient Chinese were using massive pieces of equipment called trip hammers to process food, crush bamboo to make paper, or shape red-hot metal. Too heavy to be lifted by a single person, a trip hammer has to be raised by mechanical means. The earliest of these devices were powered by waterwheels.

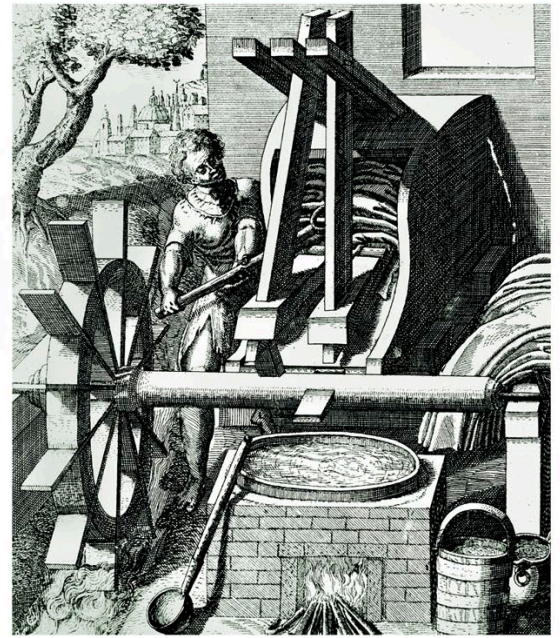


NORIA

During the Middle Ages, Arab engineers invented a type of waterwheel known as the noria. A noria raised water from streams or lakes so that it could be carried into homes for drinking or washing, or to farmland for watering crops. As the wheel turned, hollow chambers around the rim scooped up the water and emptied it into a trough to be piped elsewhere.



Water-powered cloth mill, 17th-century print



POWERING MILLS

An important use of waterwheels was in the making of cloth. The power of the turning wheel drove machinery that could spin thread and weave and finish materials.

Before the invention of coal-powered engines in the 18th century, cloth mills had to be located near running water.

HOW WATERWHEELS WORK

Blades or buckets are positioned around the rim of a waterwheel. As falling or flowing water strikes them, it turns the wheel. The rotation turns an axle at the center of the wheel that is linked to machinery.

